

# A Density-Zero Kummer Obstruction Theorem for Erdős–Graham Problem #203

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## Abstract

This draft formulates a safe Erdős–Graham #203-adjacent theorem schema extracted from the Kummer obstruction program. It does not claim a full solution of Problem #203. Instead, it isolates the exact local subgroup obstruction

$$\Gamma_d = \langle 2, 3 \rangle \subset (\mathbb{Z}/d\mathbb{Z})^*$$

and records a density-zero theorem template: the centered local obstruction has exact second moments over  $m$ , and after inserting subgroup-size lower bounds of Pappalardi type and a standard second-moment sieve, the possible  $m$  for which the Kummer local obstruction dominates form a density-zero family. The final submission must make the sieve functional and exceptional set fully explicit; this draft supplies the theorem architecture and proof skeleton.

## 1 Honest status

This paper is not a proof that for every  $(m, 6) = 1$ ,

$$|\{(k, \ell) : m2^k3^\ell + 1 \text{ prime}\}| = \infty.$$

That full assertion remains conditional on a high- $\ell$  Kummer non-concentration theorem.

The safe theorem is weaker: for 100% of  $m$ , the exact Kummer local obstruction is variance-controlled.

## 2 Local obstruction

For squarefree  $d$ , set

$$G_d = (\mathbb{Z}/d\mathbb{Z})^*, \quad \Gamma_d = \langle 2, 3 \rangle \subset G_d.$$

For  $m \in G_d$ , define

$$\rho_m(d) = \frac{\mathbf{1}_{\Gamma_d}(-m^{-1})}{|\Gamma_d|}$$

and

$$\Delta_m(d) = \rho_m(d) - \frac{1}{\varphi(d)}.$$

**Lemma 1** (Exact moments). *For squarefree  $d$  with  $(d, 6) = 1$ ,*

$$\sum_{m \in G_d} \Delta_m(d) = 0,$$

and

$$\sum_{m \in G_d} |\Delta_m(d)|^2 = \frac{1}{|\Gamma_d|} - \frac{1}{\varphi(d)}.$$

*Proof.* The map  $m \mapsto -m^{-1}$  is a bijection of  $G_d$ . The result is the variance identity for the normalized indicator of the subgroup  $\Gamma_d \subset G_d$ .  $\square$

**Definition 1** (Kummer obstruction energy). *Define*

$$\mathcal{E}(d) = \frac{1}{|\Gamma_d|} - \frac{1}{\varphi(d)}.$$

Then Lemma 1 says

$$\mathcal{E}(d) = \sum_{m \in G_d} |\Delta_m(d)|^2.$$

### 3 Prime-level weighted variance

For a finite set of primes  $\mathcal{Q}$ , let  $P = \prod_{q \in \mathcal{Q}} q$ .

**Lemma 2** (Weighted variance). *For weights  $\lambda_q$ ,*

$$\mathbb{E}_{m \in (\mathbb{Z}/P\mathbb{Z})^*} \left| \sum_{q \in \mathcal{Q}} \lambda_q \Delta_m(q) \right|^2 = \sum_{q \in \mathcal{Q}} |\lambda_q|^2 \frac{1}{q-1} \left( \frac{1}{|\Gamma_q|} - \frac{1}{q-1} \right).$$

*Proof.* Expanding the square, diagonal terms are Lemma 1. Cross terms vanish by the Chinese remainder theorem and the mean-zero identity.  $\square$

### 4 Subgroup-size input

The required arithmetic input is a lower bound for

$$|\Gamma_q| = |\langle 2, 3 \rangle \bmod q|$$

for all but a controlled set of primes. A theorem of Pappalardi type gives lower bounds for the order of subgroups generated by multiplicatively independent rationals.

**Hypothesis 1** (Pappalardi-type order input). *There exist constants  $\eta > 0$ ,  $C_1, C_2 > 0$ , and an exceptional prime set  $\mathcal{E}_{\text{ord}}(X)$  satisfying*

$$|\mathcal{E}_{\text{ord}}(X)| \leq C_1 \frac{X}{(\log X)^{1+C_2}}$$

*or another summable density bound, such that for primes  $q \leq X$ ,  $q \notin \mathcal{E}_{\text{ord}}(X)$ ,*

$$|\langle 2, 3 \rangle \bmod q| \geq q^\eta.$$

**Remark 1.** *The final paper must replace Hypothesis 1 by the exact quoted theorem, including constants and the precise exceptional-set statement.*

## 5 Density-zero theorem template

Let  $M \rightarrow \infty$ , and let  $\mathcal{Q} = \mathcal{Q}(M)$  be a prime set with associated weights  $\lambda_q$ . Define the obstruction sum

$$\mathcal{D}_m(\mathcal{Q}) = \sum_{q \in \mathcal{Q}} \lambda_q \Delta_m(q).$$

**Proposition 1** (Variance control). *Assuming the order input above and suitable support conditions on  $\lambda_q$ ,*

$$\mathbb{E}_{m \leq M, (m, 6P)=1} |\mathcal{D}_m(\mathcal{Q})|^2 = o(1)$$

*after normalization by the relevant main term.*

*Proof skeleton.* Reduce the  $m \leq M$  average to residue classes modulo  $P$ , with an error controlled by the support size. Apply Lemma 2. Use

$$\frac{1}{|\Gamma_q|} - \frac{1}{q-1} \leq \frac{1}{|\Gamma_q|}$$

and the Pappalardi-type lower bound outside the exceptional prime set. Choose  $\mathcal{Q}$  and  $\lambda_q$  so that the resulting diagonal sum is  $o(1)$  after normalization.  $\square$

**Theorem 1** (Density-zero Kummer obstruction theorem schema). *Let  $E(M)$  be the set of  $m \leq M$ ,  $(m, 6) = 1$ , for which the Kummer local obstruction sum exceeds the second-moment sieve threshold. After the explicit Pappalardi input, sieve weights, support level, and normalization are specified as in Proposition 1,*

$$|E(M)| = o(M).$$

*Proof skeleton.* Apply Proposition 1 and Chebyshev's inequality to the normalized obstruction sum. The resulting exceptional set has size  $o(M)$ . The threshold and normalization must be specified in the final paper so that the theorem states exactly what the proof controls.  $\square$

## 6 Dependency status

This draft depends on two non-elementary steps:

1. an explicit subgroup-order theorem for  $|\langle 2, 3 \rangle \bmod q|$ , preferably in the form of Pappalardi's theorem or a directly quoted variant;
2. a fully specified second-moment sieve functional.

Until these are inserted, the document should be treated as a theorem schema, not a submission-ready theorem.

## 7 Best-safe theorem posture

The strongest safe version of this paper should be titled as a theorem about the obstruction functional, not as a theorem about the full prime sequence. A submission-ready statement should have the form:

For a specified prime set  $\mathcal{Q}(M)$ , weights  $\lambda_q$ , and normalization  $V(M)$ , one has

$$\# \left\{ m \leq M : (m, 6) = 1, \left| \sum_{q \in \mathcal{Q}(M)} \lambda_q \Delta_m(q) \right| > \varepsilon V(M) \right\} = o(M).$$

This is stronger and safer than informal language about “almost all  $m$  satisfying Erdős–Graham #203.” The latter may require additional prime-producing input not present in the obstruction-energy calculation alone.

**Warning 1.** *The final theorem should not use the phrase “prime-producing mechanism” unless that mechanism is defined as a specific weighted sieve functional.*

## 8 Submission requirements

Before submission, this draft requires the following replacements:

1. Replace Hypothesis 1 with the exact Pappalardi theorem statement, including exceptional-prime bounds.
2. Define the exact sieve weights  $\lambda_q$ , prime set  $\mathcal{Q}(M)$ , and support level.
3. State the exceptional set  $E(M)$  in terms of the normalized obstruction functional actually controlled.
4. Verify the passage from residue-class averaging modulo  $P$  to integer averaging  $m \leq M$ .
5. Remove all placeholder language from Theorem 1.

## 9 What this theorem does and does not say

It does say:

the exact Kummer subgroup obstruction is density-zero exceptional in  $m$ .

It does not say:

Erdős–Graham #203 is solved for every  $m$ .

The full statement for every  $m$  requires a separate Kummer prime-distribution theorem.

## References

- [1] H. Iwaniec and E. Kowalski. *Analytic Number Theory*. AMS Colloquium Publications, 2004.
- [2] F. Pappalardi. On the order of finitely generated subgroups of  $\mathbb{Q}^*$  modulo  $p$  and divisors of  $p - 1$ . *Journal of Number Theory*, 57(2):207–222, 1996.